

Maven Ansible Web App

Step 1: Create Maven Ansible Web App using maven-archetype-webapp template.

`mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false`

```
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 26 MB/s)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/shraavan-hegde/Devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MavenAnsibleWebApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shraavan-hegde/Devops/MavenAnsibleWebApp
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.734 s
[INFO] Finished at: 2026-05-09T15:41:29+05:30
[INFO] -----
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops$ cd MavenAnsibleWebApp/
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ gedit pom.xml
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ mkdir ansible
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ ls
ansible  pom.xml  src
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ ls -la
.  ..  ansible  pom.xml  src
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ cd ansible/
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp/ansible$ gedit hosts.ini
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp/ansible$ ls
hosts.ini
shraavan-hegde@shraavan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp/ansible$ gedit playbook.yml

[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shraavan-hegde/Devops/MavenAnsibleWebApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MavenAnsibleWebApp1 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MavenAnsibleWebApp1 ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MavenAnsibleWebApp1] in [/home/shraavan-hegde/Devops/MavenAnsibleWebApp/target/MavenAnsibleWebApp1]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/shraavan-hegde/Devops/MavenAnsibleWebApp/src/main/webapp]
[INFO] Building war: /home/shraavan-hegde/Devops/MavenAnsibleWebApp/target/MavenAnsibleWebApp1.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MavenAnsibleWebApp1 ---
[INFO] Installing /home/shraavan-hegde/Devops/MavenAnsibleWebApp/target/MavenAnsibleWebApp1.war to /home/shraavan-hegde/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.war
[INFO] Installing /home/shraavan-hegde/Devops/MavenAnsibleWebApp/pom.xml to /home/shraavan-hegde/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.321 s
[INFO] Finished at: 2026-05-09T15:57:30+05:30
[INFO] -----
```

Step 2 : Edit the pom.xml file

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="
    http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/mavenv4_0_0.xsd">

  <!-- Project Model Version -->
  <modelVersion>4.0.0</modelVersion>

  <!-- Project Details -->
  <groupId>com.example</groupId>
  <artifactId>MavenAnsibleWebApp1</artifactId>
  <version>1.0-SNAPSHOT</version>

  <!-- Packaging Type -->
  <packaging>war</packaging>

  <!-- Project Name -->
  <name>MavenAnsibleWebApp1</name>

  <!-- Project URL -->
  <url>http://maven.apache.org</url>

  <!-- Dependencies -->
  <dependencies>

    <!-- JUnit Dependency for Testing -->
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>

    <!-- Servlet API for Java Web Applications -->
    <dependency>
      <groupId>javax.servlet</groupId>
      <artifactId>javax.servlet-api</artifactId>
      <version>4.0.1</version>
      <scope>provided</scope>
    </dependency>

  </dependencies>

  <!-- Build Configuration -->
  <build>
```

```

<!-- Plugins -->
<plugins>

  <!-- Maven WAR Plugin -->
  <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-war-plugin</artifactId>
    <version>3.4.0</version>
  </plugin>

</plugins>

<!-- Final WAR File Name -->
<finalName>MavenAnsibleWebApp1</finalName>

</build>

</project>

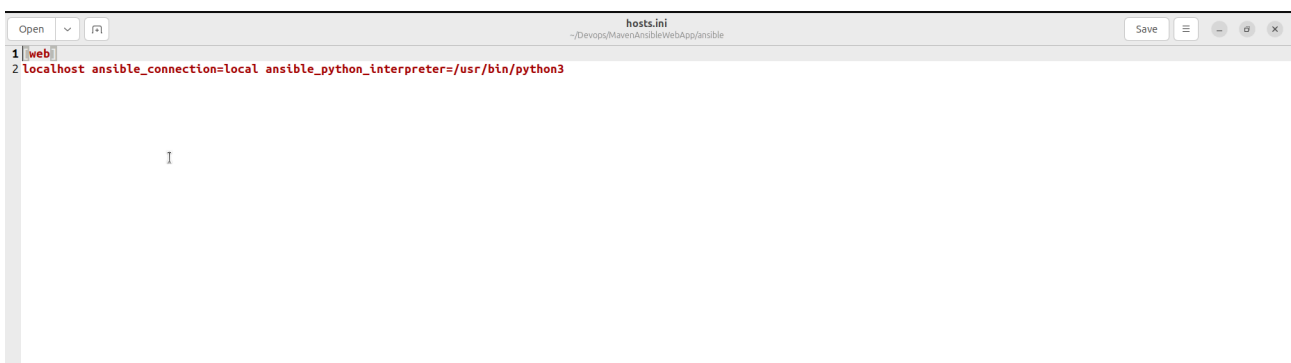
```

Step 3 : Create hosts.ini file inside ansible folder.

```

shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ cd ansible/
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp/ansible$ gedit hosts.ini
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp/ansible$ ls
hosts.ini

```



Step 4 : Create playbook.yml file inside the ansible folder.

Write the below code inside the playbook.yml file

```
1 ---
2 - name: Deploy to Tomcat
3   hosts: web
4   become: yes
5
6   tasks:
7     - name: Copy WAR file to Tomcat
8       copy:
9         src: "/var/snap/jenkins/5015/workspace/MavenAnsibleWebApp1/target/MavenAnsibleWebApp1.war"
10        dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"
11        remote_src: no
12
13    - name: Restart Tomcat
14      shell: |
15        /opt/tomcat/bin/shutdown.sh || true
16        sleep 5
17        /opt/tomcat/bin/startup.sh
18
```

Step 5 : Build the Maven Ansible Web Application.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MavenAnsibleWebApp1 >-----
[INFO] Building MavenAnsibleWebApp1 1.0-SNAPSHOT
[INFO] -----[ war ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MavenAnsibleWebApp1 ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MavenAnsibleWebApp1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MavenAnsibleWebApp1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shravan-hegde/Devops/MavenAnsibleWebApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MavenAnsibleWebApp1 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MavenAnsibleWebApp1 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MavenAnsibleWebApp1 ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MavenAnsibleWebApp1] in [/home/shravan-hegde/Devops/MavenAnsibleWebApp/target/MavenAnsibleWebApp1]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/shravan-hegde/Devops/MavenAnsibleWebApp/src/main/webapp]
[INFO] Building war: /home/shravan-hegde/Devops/MavenAnsibleWebApp/target/MavenAnsibleWebApp1.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MavenAnsibleWebApp1 ---
[INFO] Installing /home/shravan-hegde/Devops/MavenAnsibleWebApp/target/MavenAnsibleWebApp1.war to /home/shravan-hegde/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.war
[INFO] Installing /home/shravan-hegde/Devops/MavenAnsibleWebApp/pom.xml to /home/shravan-hegde/.m2/repository/com/example/MavenAnsibleWebApp1/1.0-SNAPSHOT/MavenAnsibleWebApp1-1.0-SNAPSHOT.pom
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.321 s
[INFO] Finished at: 2026-05-09T15:57:30+05:30
[INFO]
```

Step 6 : Verify the project using tree command.

Here we can see the MavenAnsibleWebApp1.war file ready for deployment.

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ tree
.
├── ansible
│   ├── hosts.ini
│   └── playbook.yml
├── pom.xml
├── src
│   ├── main
│   │   ├── resources
│   │   │   └── webapp
│   │   │       ├── index.jsp
│   │   │       └── WEB-INF
│   │   │           └── web.xml
│   └── target
│       ├── classes
│       │   ├── MavenAnsibleWebApp1
│       │   │   ├── index.jsp
│       │   │   ├── META-INF
│       │   │   ├── WEB-INF
│       │   │   │   ├── classes
│       │   │   │   └── web.xml
│       │   └── MavenAnsibleWebApp1.war
│       ├── maven-archiver
│       └── pom.properties
└── 14 directories, 9 files
```

Setting Up Jenkins Pipeline for MavenAnsibleWebApp

Step 1 : To deploy MavenAnsibleWebApp, Create the Jenkinsfile inside the project directory and write the code for deployment.

Then commit the changes and push that to the GitHub repository

```
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ gedit Jenkinsfile
shravan-hegde@shravan-hegde-QEMU-Virtual-Machine:~/Devops/MavenAnsibleWebApp$ ls
ansible Jenkinsfile pom.xml src target
```

```
pipeline {
```

```
    agent any // Use any available Jenkins agent
```

```
    environment {
        LANG = 'en_US.UTF-8'
        LC_ALL = 'en_US.UTF-8'
    }
```

```
    /*
    Add the above environment variables only if you get
    UTF-8 related errors in Jenkins.
    */
```

```
    tools {
        maven 'Maven'
        // Ensure this matches the Maven tool name configured in Jenkins
    }
```

```
    stages {
```

```

/* ----- CHECKOUT STAGE ----- */

stage('Checkout') {
    steps {
        git branch: 'master',
            url: 'https://github.com/ShruthiBGowda/MavenAnsibleWebApp1CICD.git'
    }
}

/* ----- BUILD STAGE ----- */

stage('Build') {
    steps {
        sh 'mvn clean package'
        // Builds the Maven project and generates WAR file
    }
}

/* ----- ARCHIVE STAGE ----- */

stage('Archive') {
    steps {
        archiveArtifacts artifacts: 'target/*.war',
            fingerprint: true
    }
}

/* ----- DEPLOY STAGE ----- */

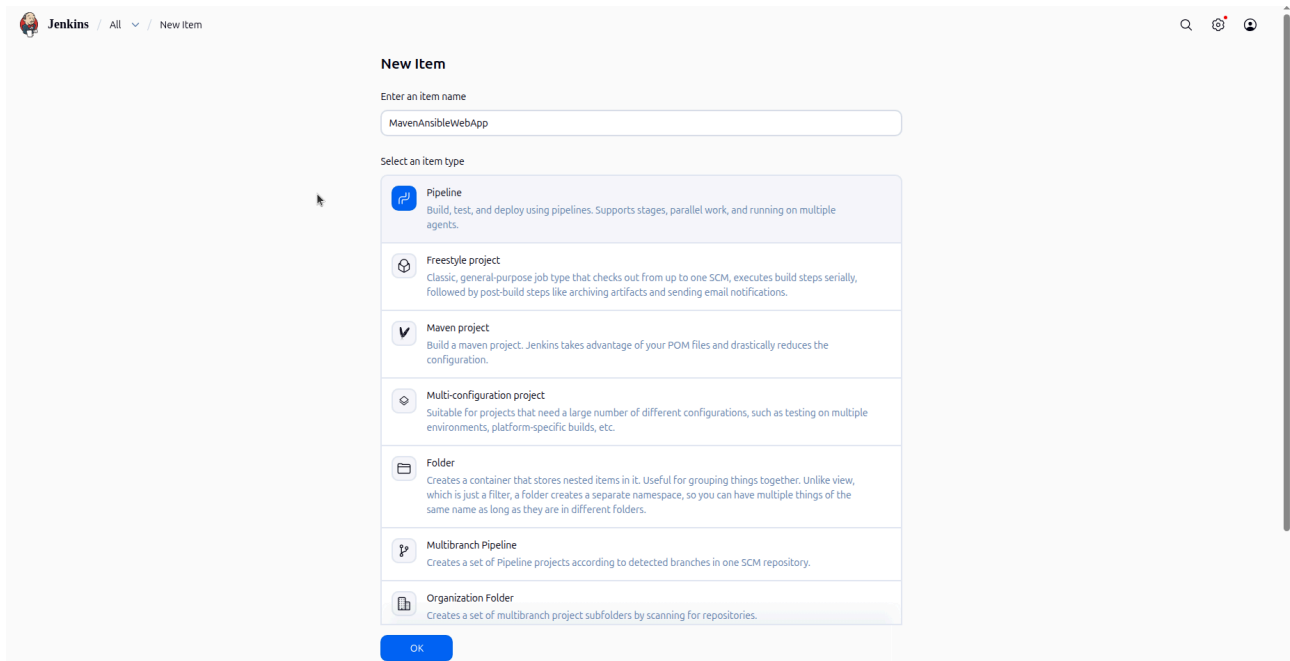
stage('Deploy') {
    steps {

        // Build project again before deployment
        sh 'mvn clean package'

        // Run Ansible Playbook for deployment
        sh 'ansible-playbook ansible/playbook.yml -i ansible/hosts.ini'
    }
}
}
}

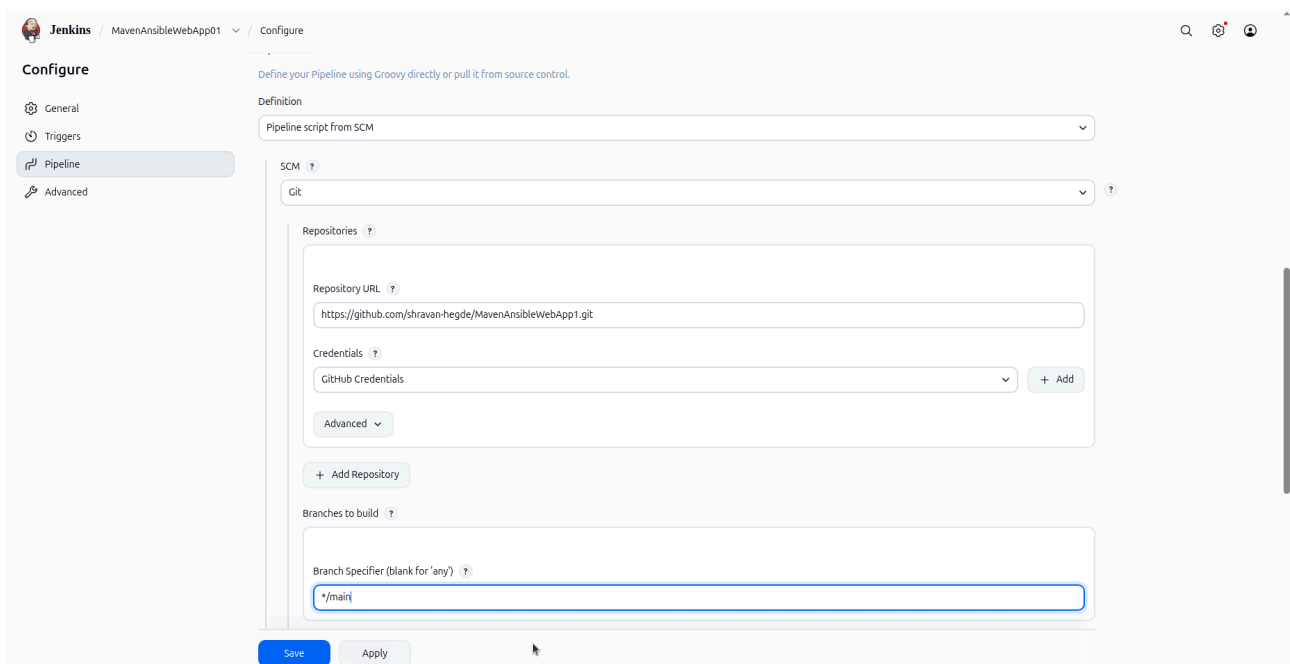
```

Step 2 : Click on New Item in order for creating an New Item and thus give the name as “MavenAnsibleWebApp1” and Item type as Pipeline and



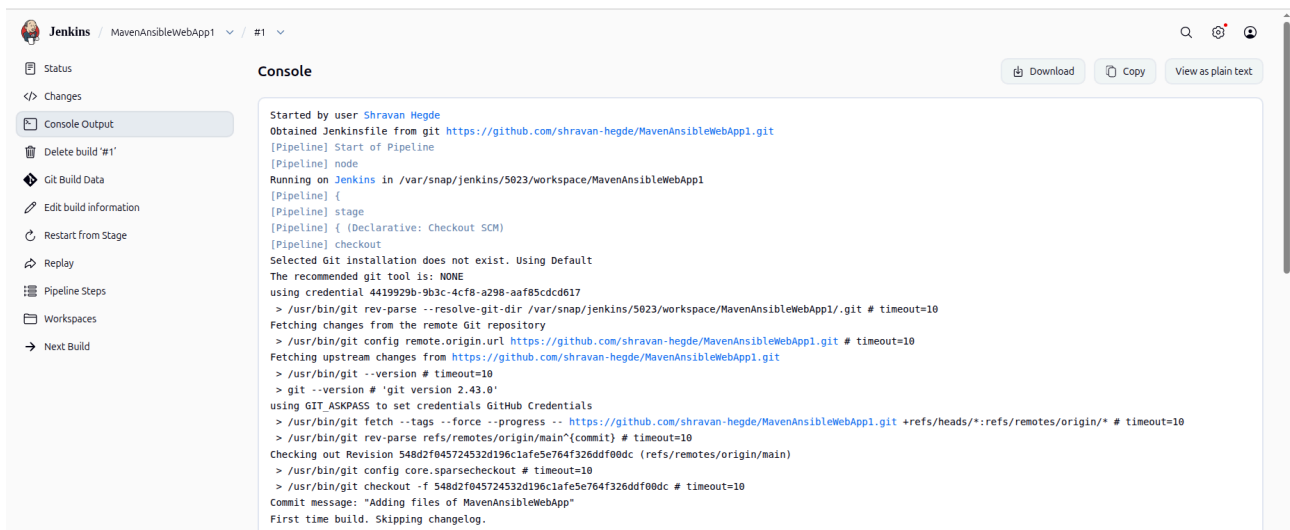
The screenshot shows the Jenkins 'New Item' dialog. At the top, the breadcrumb is 'Jenkins / All / New Item'. The 'Enter an item name' field contains 'MavenAnsibleWebApp'. Below it, the 'Select an item type' section lists several options: Pipeline (selected with a blue icon), Freestyle project, Maven project (checked with a green checkmark), Multi-configuration project, Folder, Multibranch Pipeline, and Organization Folder. Each option has a brief description. At the bottom of the dialog is an 'OK' button.

Step 3 : To configure the pipeline, set the Definition to Pipeline script from SCM, choose Git as your SCM, and provide your Repository URL along with the GitHub Credentials you previously established.



The screenshot shows the Jenkins 'Configure' page for the item 'MavenAnsibleWebApp01'. The left sidebar has tabs for 'General', 'Triggers', 'Pipeline', and 'Advanced', with 'Pipeline' currently selected. The main content area is titled 'Define your Pipeline using Groovy directly or pull it from source control.' Under the 'Definition' section, 'Pipeline script from SCM' is selected. Below this, the 'SCM' dropdown is set to 'Git'. The 'Repositories' section contains a 'Repository URL' field with the value 'https://github.com/shravan-hegde/MavenAnsibleWebApp1.git', a 'Credentials' dropdown set to 'GitHub Credentials', and an 'Advanced' dropdown. There is an '+ Add Repository' button. The 'Branches to build' section has a 'Branch Specifier (blank for \'any\')' field with the value '*/main'. At the bottom are 'Save' and 'Apply' buttons.

Step 4 : Build and TroubleShoot Deployment in Jenkins.



Step 5 : Change the playbook.yml file to resolve the error.

```
- name: Deploy to Tomcat
  hosts: web
  become: yes
```

tasks:

```
# Copy WAR file to Tomcat webapps folder
```

```
- name: Copy WAR file to Tomcat
```

```
  copy:
```

```
    src: "/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1CICD/target/
```

```
MavenAnsibleWebApp1.war"
```

```
    dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"
```

```
    remote_src: no
```

```
# Restart Tomcat Server
```

```
- name: Restart Tomcat
```

```
  shell: |
```

```
    /opt/tomcat/bin/shutdown.sh || true
```

```
    sleep 5
```

```
    /opt/tomcat/bin/startup.sh
```


Step 6: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.

The screenshot shows the Jenkins interface for the **MavenAnsibleWebApp1** pipeline. The left sidebar contains navigation options: Status, Changes, Build Now, Delete Pipeline, Configure, Rename, Full Stage View, and Pipeline Syntax. The main area displays the **Stage View** for the pipeline, showing a table of build results for various stages across multiple builds.

Stage View Table:

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Archive	Deploy
Average stage times: (full run time: ~18s)	1s	83ms	881ms	1s	135ms	5s
#6 May 09 16:55 No Changes	849ms	76ms	879ms	2s	179ms	10s
#5 May 09 16:50 No Changes	823ms	60ms	823ms	2s	163ms	11s
#4 May 09 16:45 1 commit	1s	113ms	847ms	2s	181ms	10s
#3 May 09 16:38 1 commit	1s	59ms	906ms	3s	202ms	2s
#2 May 09 16:29 1 commit	1s	72ms	885ms	46ms	41ms	34ms
#1 May 09 16:24 No Changes	2s	118ms	950ms	92ms	45ms	45ms

The table indicates that builds #1 through #4 were successful, while builds #2, #3, and #5 failed in the Deploy stage. The 'failed' status is indicated by a red background and the word 'failed' in the Deploy column.